

Lecture 11: Tangent Planes & Linear Approximations

Background & Motivation

In Calc I, the tangent line at a point gives the best linear approximation: $f(x) \approx f(a) + f'(a)(x - a)$. Today we extend this to surfaces: the tangent plane is the best flat approximation to $z = f(x, y)$ near a point. This is the foundation of linearization, differentials, and error estimation in multiple dimensions.

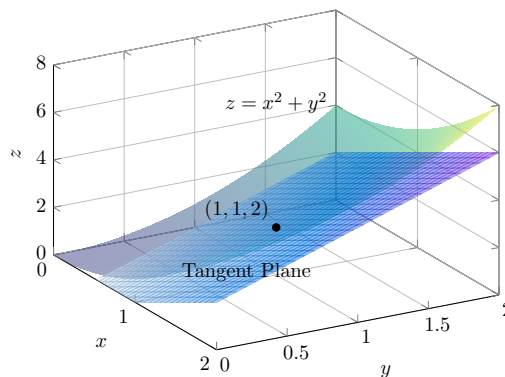
1. From Tangent Lines to Tangent Planes

- In Calc I: tangent line at $x = a$ gives $f(x) \approx f(a) + f'(a)(x - a)$
- In Calc III: the tangent *plane* at (a, b) is the best linear approximation to a surface

Theorem 1: Equation of the Tangent Plane

The tangent plane to $z = f(x, y)$ at the point $(a, b, f(a, b))$:

$$z = \underline{\hspace{2cm}}$$



Example 1: Tangent Plane to a Paraboloid

Find the tangent plane to $z = x^2 + y^2$ at $(1, 1, 2)$.

1. **Compute partials:**

$$f_x = 2x, \quad f_y = 2y$$

2. **Evaluate at $(1, 1)$:**

$$f_x(1, 1) = \underline{\hspace{2cm}}, \quad f_y(1, 1) = \underline{\hspace{2cm}}, \quad f(1, 1) = \underline{\hspace{2cm}}$$

3. **Tangent plane:**

Example 2: Tangent Plane to $z = y^4 + xy - 2y$ at $(1, 1)$

Find the equation of the tangent plane.

1. **Function value:** $f(1, 1) = 1 + 1 - 2 =$ _____

2. **Partials:**

3. **Result:** $z = 0 + 1(x - 1) + 3(y - 1) =$ _____

2. Differentiability

Definition 1: Differentiability in Several Variables

$f(x, y)$ is **differentiable** at (a, b) if:

$$f(a + \Delta x, b + \Delta y) = f(a, b) + f_x \Delta x + f_y \Delta y + E(\Delta x, \Delta y)$$

where the error E satisfies $\frac{E}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$.

- If f_x and f_y are **continuous** near (a, b) , then f is differentiable there
- Differentiable $\implies$ continuous (but not the other way around)

3. Linear Approximation

Linear Approximation Formula

$$f(x, y) \approx L(x, y) =$$

- The tangent plane *is* the linear approximation
- Accuracy improves as (x, y) gets closer to (a, b)

Example 3: Practical Linear Approximation

Use linear approximation to estimate $f(2.1, 0.9)$ where $f(x, y) = x^2y - xy^3$.

1. **Base point:** $(a, b) = (2, 1)$, $f(2, 1) = 4 - 2 =$ _____

2. **Partials at base point:**

3. **Approximate:**

$$L(2.1, 0.9) = 2 + 3(0.1) + (-2)(-0.1) = 2 + 0.3 + 0.2 =$$

Check: $f(2.1, 0.9) = (4.41)(0.9) - (2.1)(0.729) = 3.969 - 1.531 = 2.438$. Error ≈ 0.06 .

4. Tangent Planes to Implicit Surfaces

– For a surface given implicitly as $F(x, y, z) = k$, the gradient ∇F is normal to the surface

Tangent Plane to $F(x, y, z) = k$ at (x_0, y_0, z_0)

Equivalently: $\nabla F \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$

Example 4: Tangent Plane to a Sphere

Find the tangent plane to $x^2 + y^2 + z^2 = 14$ at $(1, 2, 3)$.

1. **Set up:** $F(x, y, z) = x^2 + y^2 + z^2$
2. **Gradient:** $\nabla F = \langle 2x, 2y, 2z \rangle$. At $(1, 2, 3)$: $\nabla F =$ _____
3. **Tangent plane:**

Practice Problems

Problem 1. Find the tangent plane to $z = xe^y$ at $(2, 0, 2)$.

Problem 2. Use linear approximation to estimate $\sqrt{(3.02)^2 + (3.97)^2}$.

Hint: Let $f(x, y) = \sqrt{x^2 + y^2}$ with base point $(3, 4)$.

Problem 3. Find the tangent plane to $x^2 + 2y^2 + 3z^2 = 36$ at $(1, 2, 3)$.

Problem 4. Find the tangent plane to $z = \sin(xy)$ at $(\pi/2, 1, 1)$ and use it to approximate $f(1.5, 1.1)$.

Problem 5. Two surfaces $z = x^2 + y^2$ and $z = 4 - x^2 - y^2$ intersect along a curve. Find the tangent line to this curve at $(1, 1, 2)$.